

ECON 636 600: MACROECONOMIC THEORY I

Midterm (Oct 24)

1. (45 points) Consider a two period growth model. $u(c) = \log(c)$ and $f(k) = k^\alpha$, $\delta = 1$ and $t = 0, 1$. The planner chooses $\{c_0, k_1, c_1\}$ to maximize

$$\max \quad \log(c_0) + \beta \log(c_1)$$

$$s.t. \quad c_0 + k_1 = k_0^\alpha$$

$$c_1 = k_1^\alpha$$

$$k_0 \text{ is given.}$$

- (a) Write down the Euler equation.

$$u'(c_0) = \beta u'(c_1) \alpha k_1^{\alpha-1}$$

- (b) Solve for the optimal c_0, k_1, c_1 .

Euler equation implies

$$\frac{1}{c_0} = \frac{\beta}{k_1^\alpha} \alpha k_1^{\alpha-1}$$

plug it into the budget constraint $c_0 + k_1 = k_0^\alpha$, we get

$$c_0 = \frac{1}{1 + \alpha\beta} k_0^\alpha$$

$$k_1 = \frac{\alpha\beta}{1 + \alpha\beta} k_0^\alpha$$

$$c_1 = \left(\frac{\alpha\beta}{1 + \alpha\beta} k_0^\alpha \right)^\alpha$$

2. (40 points) Consider equation $x^2 = 0$, which has a unique root $x^* = 0$.

(a) Start with initial guess $x_0 = 1$, compute x_1 and x_2 using the iteration in the newton's method.

Newton's iteration is

$$x_{k+1} = x_k - \frac{x_k^2}{2x_k} = x_k/2$$

Hence $x_1 = x_0/2 = 0.5$ and $x_2 = x_1/2 = 0.25$.

(b) Show that the sequence $\{x_k; k = 1, 2, \dots\}$ converges to the root, i.e., $\lim_{k \rightarrow \infty} x_k = 0$.

$$\lim_k x_k = \lim_k \frac{1}{2^k} = 0.$$

3. (15 points) An unemployed worker must choose the number of offers n to solicit. At a cost of $k(n)$, the worker receives n offers this period. Here $k(n+1) > k(n)$ for $n \geq 1$. The number of offers must be chosen in advance at the beginning of the period and cannot be revised during the period. The worker wants to maximize $E \sum_{t=0}^{\infty} \beta^t y_t$. Here y_t is w each period when she is employed but not searching, $[w - k(n)]$ the first period she is employed with the highest offer w but searches for n offers, and $[-k(n)]$ each period she solicits and rejects n offers. The offers are each independently drawn from $F(w)$. The worker who accepts an offer works forever at wage w .

Let V be the discounted utility of an unemployed worker who has not yet chosen the number of offers to solicit. Formulate the Functional equation that is satisfied by V .

$$V = \max_{n \geq 1} \left\{ \int_{w_1} \int_{w_2} \cdots \int_{w_n} \max_{\text{accept or reject}} \left\{ -k(n) + \frac{w}{1-\beta}, -k(n) + \beta V \right\} dF(w_1) dF(w_2) \cdots dF(w_n) \right\},$$

where $w = \max\{w_i; i = 1, \dots, n\}$.